English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, Computer Science

Test 1, 13 May 2023: solutions

English transcription of the supplied handwritten workings. The notation and derivations follow the notes, including inconsistencies identified below.

Problem 1

1. Number the prize boxes from 0 to 4. The reached box number is $Y \sim Bin(4, 1/2)$. For the payout X,

$$\begin{array}{c|ccccc} X & 1000 & 100 & 700 & 250 & 500 \\ \hline \mathbb{P} & 1/16 & 4/16 & 6/16 & 4/16 & 1/16 \end{array}$$

$$\mathbb{E}X = \frac{1000 + 400 + 4200 + 1000 + 500}{16} = \frac{7100}{16} = 443.75.$$

2. Let X_1 be the number of balls until the first 1000-lev prize. Then $X_1 \sim Ge(1/16)$ and $\mathbb{E}X_1 = 16$. Let X_2 be the total payout until then. Informally, the final 1000 leva are preceded by an average of 15 other payouts. Their conditional law is

$$\begin{array}{c|cccc} X \mid X \neq 1000 & 100 & 700 & 250 & 500 \\ \hline \mathbb{P} & 4/15 & 6/15 & 4/15 & 1/15 \end{array}$$

   so $\mathbb{E}[X \mid X \neq 1000] = 6100/15 = 406\frac{2}{3}$, yielding $1000 + 6100 = 7100$. Formally,

$$\begin{aligned} \mathbb{E}X_2 &= \sum_{k \geq 1} \mathbb{P}(X_1 = k)\mathbb{E}[X_2 \mid X_1 = k] \\ &= \sum_{k \geq 1} \left(\frac{15}{16}\right)^{k-1} \frac{1}{16} \left[(k-1)\frac{6100}{15} + 1000\right]. \end{aligned}$$

   Use

$$\sum_{k=1}^{\infty} \left(\frac{15}{16}\right)^{k-1} (k-1) = \sum_{j=0}^{\infty} j \left(\frac{15}{16}\right)^{j} = \frac{15/16}{(1/16)^2} = 15 \cdot 16$$

   and $\sum_{k \geq 1} (15/16)^{k-1}/16 = 1$. Hence $\mathbb{E}X_2 = 7100$.

3. Let Z_i be the randomly assigned prize in box i. Each is uniform on $\{1000, 100, 700, 250, 500\}$, so $\mathbb{E}Z_i = 2550/5 = 510$. Then

$$\mathbb{E}[\text{payout}] = \mathbb{E}\sum_{i=0}^{4} Z_i \binom{4}{i} (1/2)^4 = 510.$$

   The proposed best arrangement in the notes is $100, 700, 1000, 500, 250$.

4. The probability of hitting the 100-lev prize on each of three drops is

$$f(p) = (4p(1-p)^3)^3 = 64p^3(1-p)^9.$$

$$f'(p) = 192p^2(1-p)^9 - 576p^3(1-p)^8.$$

   The interior critical point satisfies $192(1-p) = 576p$, giving $p = 1/4$.

5. Bayes' rule gives

$$\frac{(3/8)^2}{(3/8)^2 + (8/27)^2} \approx 62\%.$$

Problem 2

Let $Y_1 = X_1 + X_2$ and $Y_2 = X_1 X_2$, with parameters indexed by 1 or 2 as appropriate.

1. Bernoulli variables

For iid $X_i \sim Ber(p)$, the sum takes values in $\{0,1,2\}$ and is not Bernoulli. Since $\mathbb{P}(X_1X_2 = 1) = p^2$, the product is $Ber(p^2)$.

2. Binomial variables

For iid $X_i \sim Bin(n,p)$, the sum of the two independent sums of n Bernoulli variables is $Bin(2n,p)$. If $X_1X_2 \sim Bin(n_2,p_2)$, its largest possible value forces $n_2 = n^2$. Its expectation gives

$$\mathbb{E}X_1X_2 = n^2p^2 = n_2p_2, \qquad p_2 = p^2.$$

The probability of its maximum would have to satisfy

$$p^{2n} = \mathbb{P}(X_1X_2 = n^2) = p_2^{n_2} = p^{2n^2}.$$

The notes conclude a contradiction except for the degenerate $p = 1$ case, where $X_1X_2 \sim Bin(n^2,1)$. (The $n = 1$ Bernoulli case is treated separately above.)

3. Geometric variables

The notes use the geometric convention with mean $1/p$. If $X_1 + X_2 \sim Ge(p_2)$, its mean would give $2/p = 1/p_2$, hence $p_2 = p/2$. Matching variances would require

$$\frac{2(1-p)}{p^2} = \frac{1-p_2}{p_2^2} = \frac{4-2p}{p^2},$$

which is impossible. If $X_1X_2 \sim Ge(p_3)$, the mean gives $p_3 = p^2$. On the other hand,

$$\begin{aligned} D(X_1X_2) &= \mathbb{E}X_1^2X_2^2 - (\mathbb{E}X_1X_2)^2 \\ &= \left(\frac{1-p}{p^2} + \frac{1}{p^2}\right)^2 - \left(\frac{1}{p^2}\right)^2 = \frac{3-4p+p^2}{p^4}. \end{aligned}$$

Equating this with $(1-p^2)/p^4$ gives $2p^2 - 4p + 2 = 0$, or $p = 1$. Thus only the degenerate $Ge(1)$ case works.

4. Poisson variables

For iid $X_i \sim Po(\lambda)$, $X_1 + X_2 \sim Po(2\lambda)$, for example by generating functions (see the exercises). If the product were $Po(\lambda_2)$, its mean would force $\lambda_2 = \lambda^2$, but

$$D(X_1X_2) = (\lambda + \lambda^2)^2 - \lambda^4 = \lambda^2 + 2\lambda^3.$$

Matching mean and variance would give $2\lambda^3 = 0$, a contradiction for $\lambda > 0$.

Problem 3

Write

$$f_\sigma = \tfrac{1}{4}\big(f(X_1,X_2,X_3) + f(X_2,X_1,X_3) + f(X_3,X_2,X_1) + f(X_1,X_3,X_2)\big).$$

1. Sum

For $f = X_1 + X_2 + X_3$, $\mathbb{E}f = 3\mu$, $Df = 3\sigma^2$, and $f_\sigma = f$. Thus both means and variances agree.

2. Signed sum

For $f = X_1 + X_2 - X_3$,

$$f_\sigma = \tfrac{1}{4}\big(X_1 + X_2 - X_3 + X_3 + X_2 - X_1 + X_2 + X_1 - X_3 + X_1 + X_3 - X_2\big) = \frac{X_1 + X_2}{2}.$$

Hence $\mathbb{E}f_\sigma = \mathbb{E}f = \mu$, while $Df = 3\sigma^2$ and

$$Df_\sigma = \tfrac{1}{4}(DX_1 + DX_2) = \sigma^2/2.$$

The averaging has an effect: the expectation is unchanged and the variance is smaller.

3. Product minus the third variable

The notes define

$$Y_1 = X_1X_2 - X_3, \quad Y_2 = X_3X_2 - X_1, \quad Y_3 = X_2X_3 - X_1 = Y_2, \quad Y_4 = X_1X_3 - X_2.$$

For each, $\mathbb{E}Y_i = \mu^2 - \mu$, and

$$\begin{aligned} Df = DY_1 &= D(X_1X_2) + DX_3 \\ &= (\sigma^2 + \mu^2)^2 - \mu^4 + \sigma^2 = \sigma^2(\sigma^2 + 2\mu^2 + 1). \end{aligned}$$

The subsequent handwritten calculation is

$$\begin{aligned} Df_\sigma &= \tfrac{1}{16} D(Y_1 + 2Y_2 + Y_4) \\ &= \tfrac{3}{8} DY_1 + \tfrac{5}{8} Cov(Y_1, Y_2), \\ Df - Df_\sigma &= \tfrac{5}{8}\big(DY_1 - Cov(Y_1, Y_2)\big). \end{aligned}$$

If the latter is positive, f_σ has smaller variance. Expanding the covariance gives

$$\begin{aligned} Cov(X_1X_2 - X_3, X_3X_2 - X_1) &= Cov(X_1X_2, X_2X_3) - 2Cov(X_1, X_1X_2) \\ &= \mu^2(\sigma^2 + \mu^2) - \mu^4 - 2\big((\sigma^2 + \mu^2)\mu - \mu^3\big) \\ &= \sigma^2(\mu^2 - 2\mu). \end{aligned}$$

Therefore the displayed conclusion is

$$Df - Df_\sigma = \tfrac{5}{8}\sigma^2(\sigma^2 + \mu^2 + 2\mu + 1) > 0.$$

Transcription note: this part of the handwriting duplicates Y_2 as Y_3, although the preceding definition of f_σ uses a different permutation. It also writes $\mathbb{E}f_\sigma = \mu$ on the following page, inconsistent with its earlier $\mu^2 - \mu$. These source inconsistencies have not been silently replaced by a new solution.